

PROJECT REQUIREMENTS & FACULTY PRACTICAL MATRIX

Course: Cloud Infrastructure Engineering | **Student:** 2400031215 - Bhimavarapu Hema Varshith Reddy

1. Technical Project Specifications

Requirement Category	Specification & Constraint
Target Population	50 Enrolled Engineering Students (Testing determines simultaneous user capacity)
Virtual Desktop Architecture	Pooled Host Pool, Breadth-First load balancing, Max 5 sessions per test host
Session Host Operating System	Windows 11 Enterprise Multi-Session (Build 23H2, AVD-optimized)
User Profile Management	FSLogix Profile Containers in VHDX dynamic format hosted on Azure Files
Storage Authentication	Microsoft Entra Kerberos (AADKERB) with SMB Share Contributor RBAC
Storage Performance Tier	Azure Files StandardV2_LRS, 128 GiB, 1026 IOPS provisioned, 63 MiB/s throughput
Cost & Budget Policy	Azure for Students subscription (\$100 budget cap); Automated Daily Deallocation (18:00 UTC)
Infrastructure as Code	Azure Bicep modular templates with declarative ARM integration

2. Alignment with Faculty Practical Tasks (Tasks 1–5)

The project integrates and fulfills the Azure practical curriculum specified by faculty:

Task #	Faculty Task Description	Project Implementation Details	Status
Task 1	PowerShell in Cloud Shell, Create RG, Create Cloud Storage Account, Create VM, Create Managed Disk	Created ngsg-avd-lab; configured 128 GB Standard SSD Managed Disk for session host	SATISFIED
Task 2	Entra users/groups, assigned membership, dynamic assignment, security group AVD-Engineering-Lab-Students, create user, assign role	Created Entra group AVD-Engineering-Lab-Students, created student user, assigned role	SATISFIED
Task 3	Deploy to existing RG, move resources, implement Resource Lock	Deployed resources to existing RG; applied CanNotDelete storage account.	SATISFIED
Task 4	Virtual Network, deploy VMs, private/public IP Subnets, NSG, DNS	Created vnet-avd-lab, snet-avd, deployed nsg-avd-lab with outbound rules (443, 445, 1688).	SATISFIED
Task 5	Deploy Azure VM, Storage Account, manage file shares, Entra Kerberos identity, Windows 11 host VM	Configured profiles260916, fslogix share, Entra Kerberos identity, started Windows 11 host VM	SATISFIED

3. Empirical Testing & Validation Protocol

- **Stage 1: Single User Baseline:** Verify desktop launch, measure initial logon duration, container mount time, and idle RAM/CPU footprint.
- **Stage 2: 3-User Concurrency:** Measure multi-user CPU scheduling, RAM allocation under active application launch, and storage IOPS.
- **Stage 3: 5-User Peak Load:** Push single host to designated limit, evaluate storage queue length and responsiveness, extrapolate 50-user capacity.